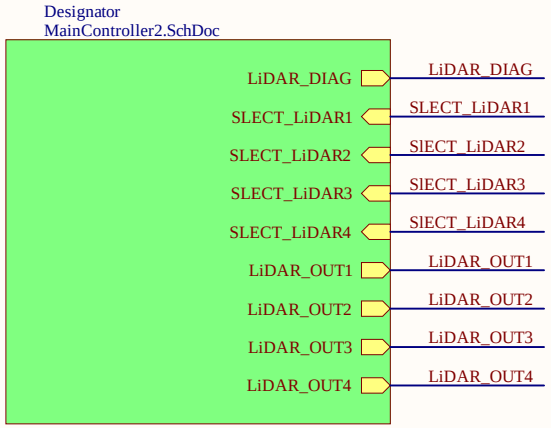
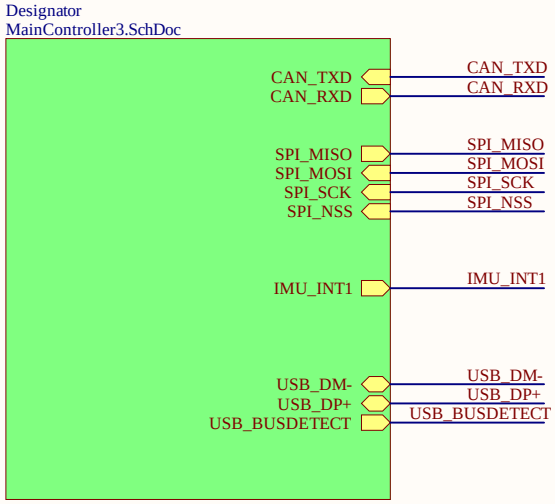
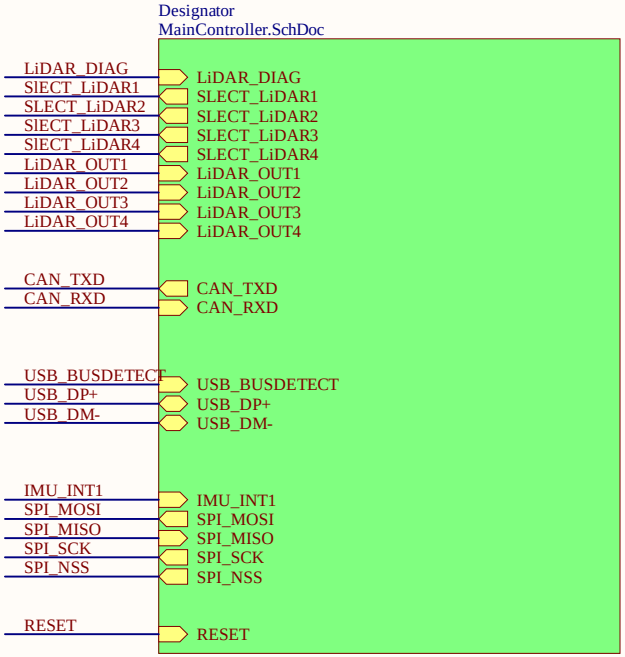
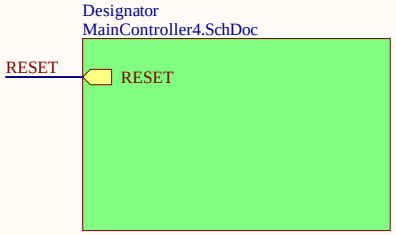
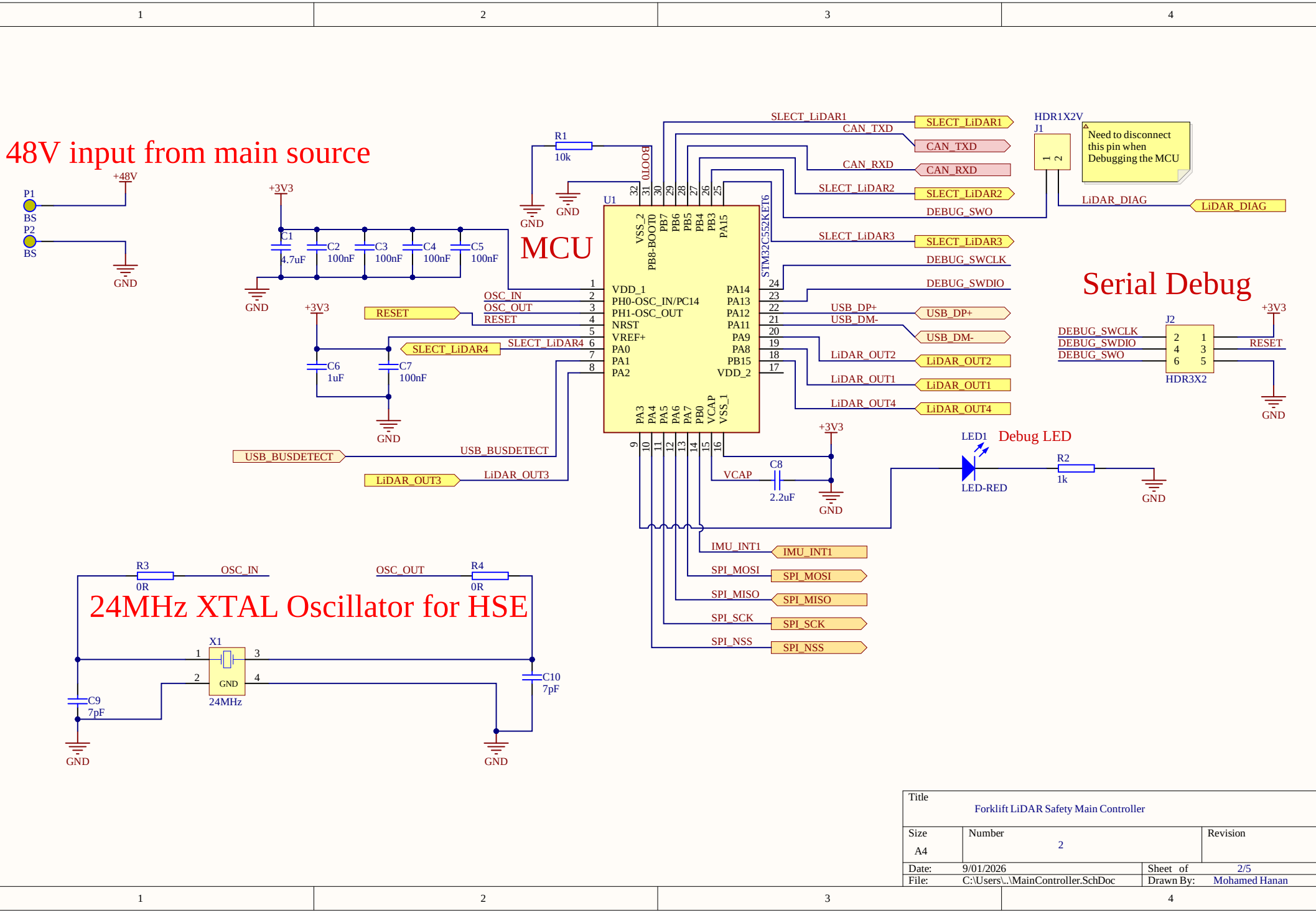






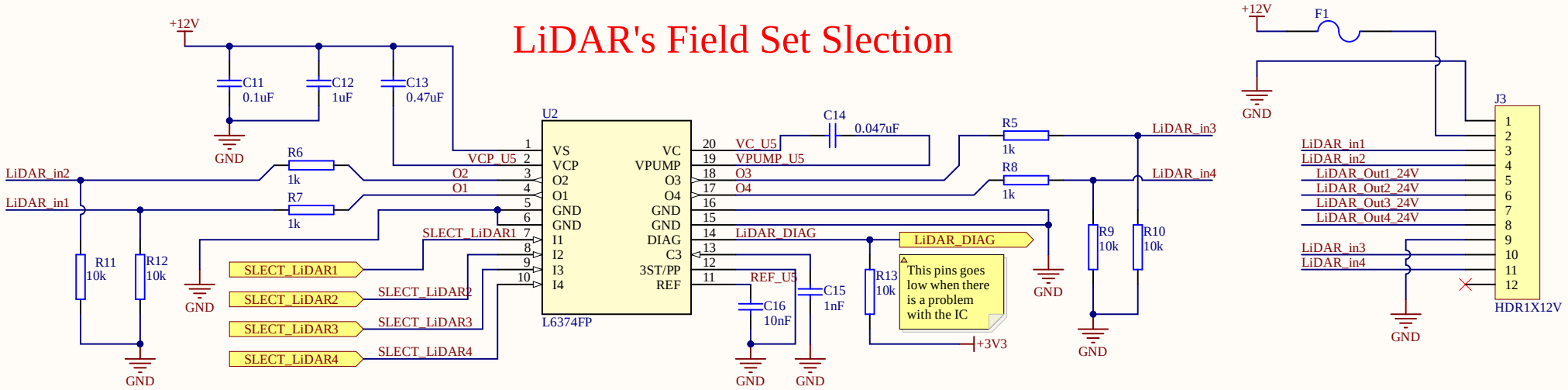


Title: Forklift LiDAR Safety Main Controller		
Size: A4	Number: 1	Revision:
Date: 9/01/2026		Sheet of 1/5
File: C:\Users\..\ConnectionMain.SchDoc		Drawn By: Mohamed Hanan

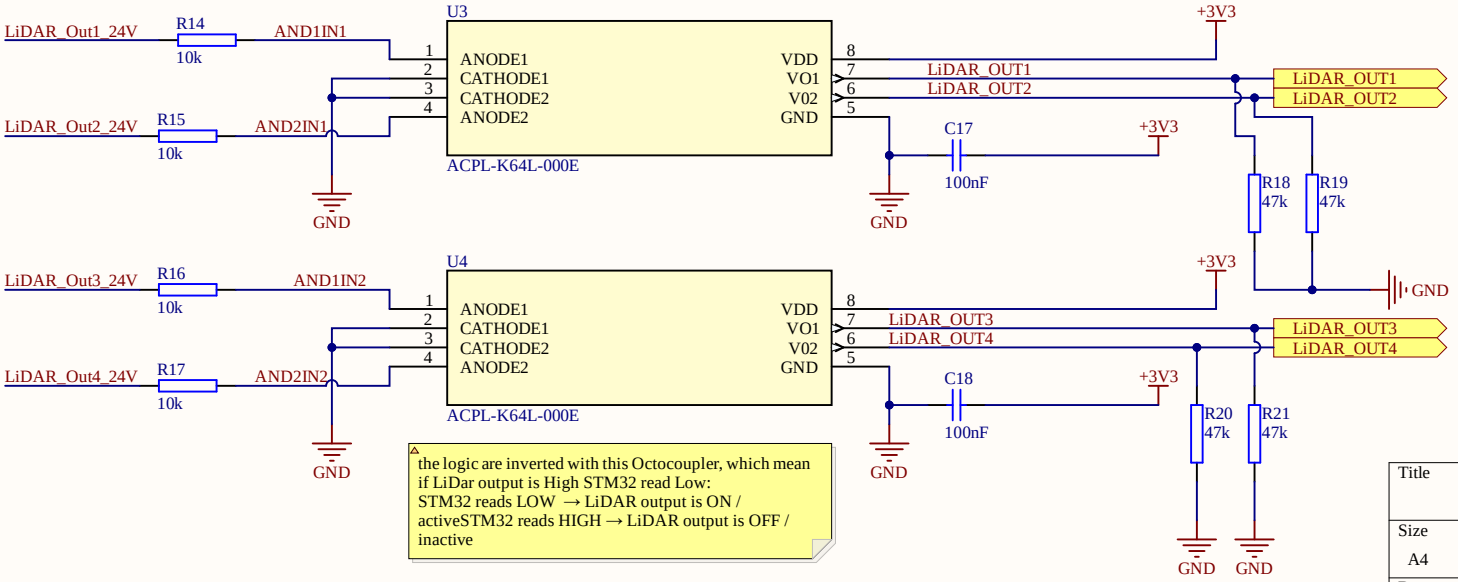


Title	Forklift LiDAR Safety Main Controller	
Size: A4	Number: 2	Revision:
Date: 9/01/2026	Sheet of 2/5	
File: C:\Users\..\MainController.SchDoc	Drawn By: Mohamed Hanan	

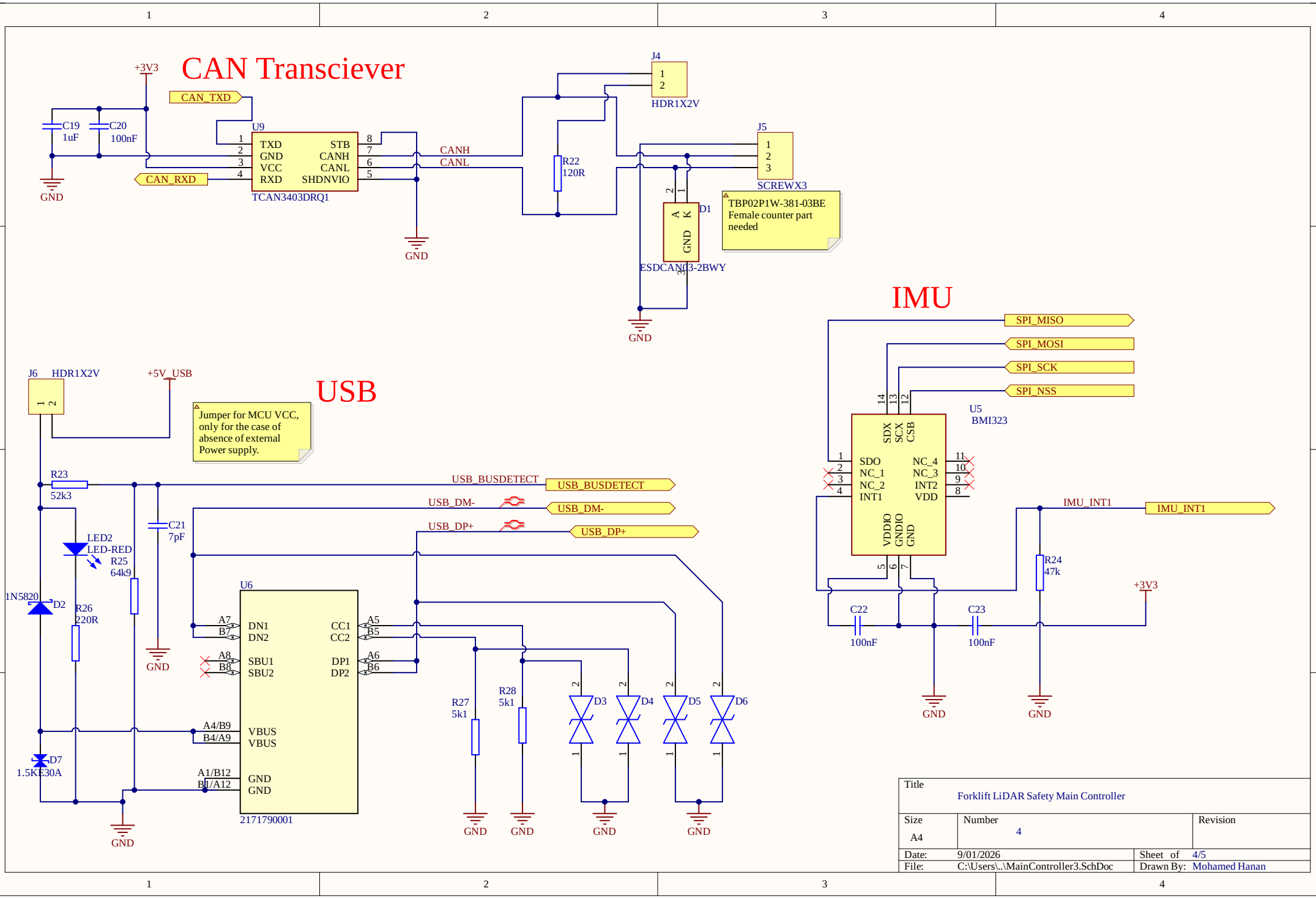
LiDAR's Field Set Slection



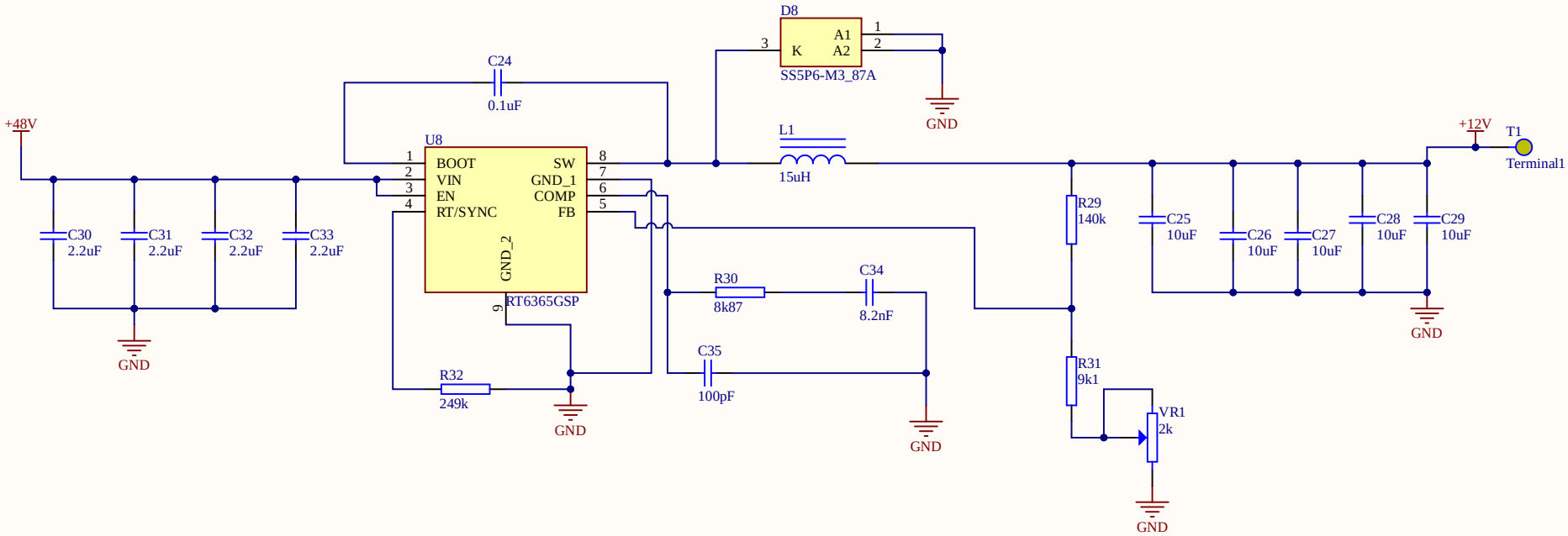
LiDAR's Logic Output



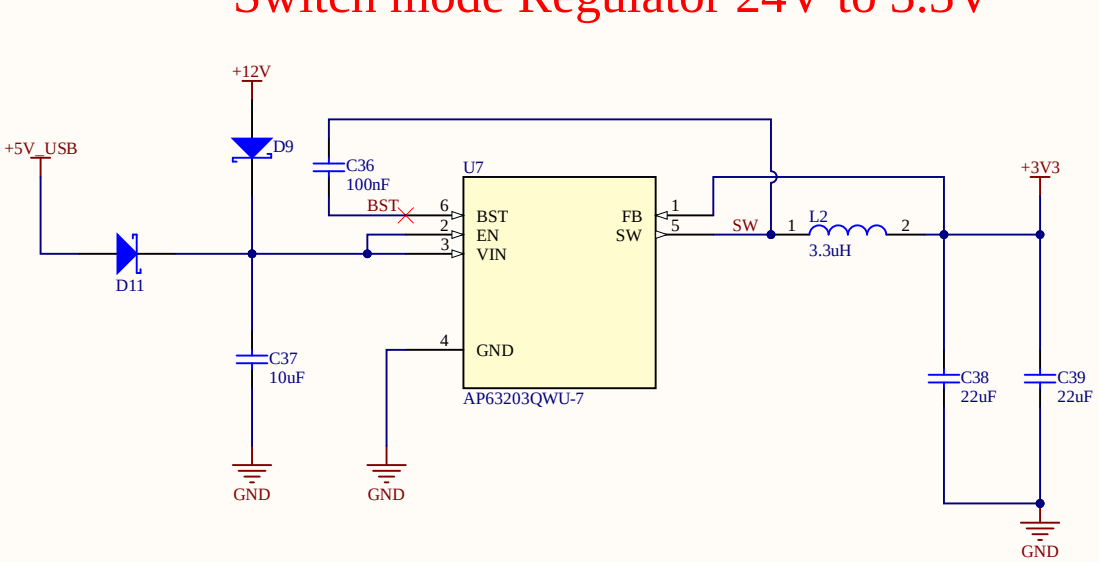
Title	Forklift LiDAR Safety Main Controller	
Size: A4	Number: 3	Revision:
Date: 9/01/2026		Sheet of 3/5
File: C:\Users\..\MainController2.SchDoc		Drawn By: Mohamed Hanan


CAN Transciever
+3V3
CAN_TXD
CAN_RXD
C19
1uF
C20
100nF
U9
TXD
GND
VCC
RXD
STB
CANH
CANL
SHDNVIO
TCAN3403DRQ1
CANH
CANL
R22
120R
J4
HDR1X2V
J5
SCREWX3
TBP02P1W-381-03BE
Female counter part
needed
D1
ESDCAN03-2BWY
GND
USB
J6
HDR1X2V
+5V_USB
Jumper for MCU VCC,
only for the case of
absence of external
Power supply.
R23
52k3
C21
7pF
LED2
LED-RED
R25
64k9
1N5820
D2
R26
220R
D7
1.5KE30A
U6
DN1
DN2
SBU1
SBU2
VBUS
VBUS
GND
GND
CC1
CC2
DP1
DP2
2171790001
USB_BUSDETECT
USB_DM-
USB_DP+
R27
5k1
R28
5k1
D3
D4
D5
D6
IMU
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_NSS
U5
BMI323
SDX
SCX
CSB
SDO
NC_1
NC_2
INT1
NC_4
NC_3
INT2
VDD
VDDIO
GNDIO
GND
IMU_INT1
R24
47k
C22
100nF
C23
100nF
+3V3
Title
Forklift LiDAR Safety Main Controller
Size
A4
Number
4
Revision
Date:
9/01/2026
Sheet of
4/5
File:
C:\Users\..\MainController3.SchDoc
Drawn By:
Mohamed Hanan

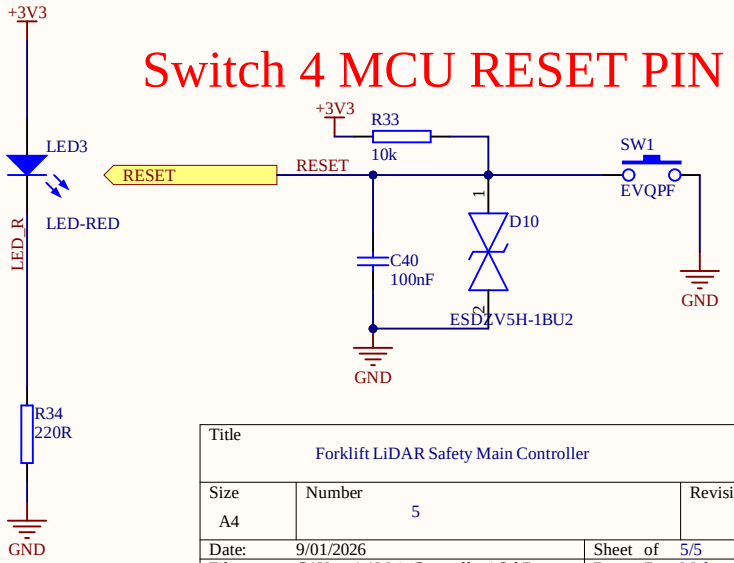
Switch mode converter form 48V to 12V



Switch mode Regulator 24V to 3.3V



Switch 4 MCU RESET PIN



Title: Forklift LiDAR Safety Main Controller		
Size: A4	Number: 5	Revision:
Date: 9/01/2026		Sheet of 5/5
File: C:\Users\..\MainController4.SchDoc		Drawn By: Mohamed Hanan